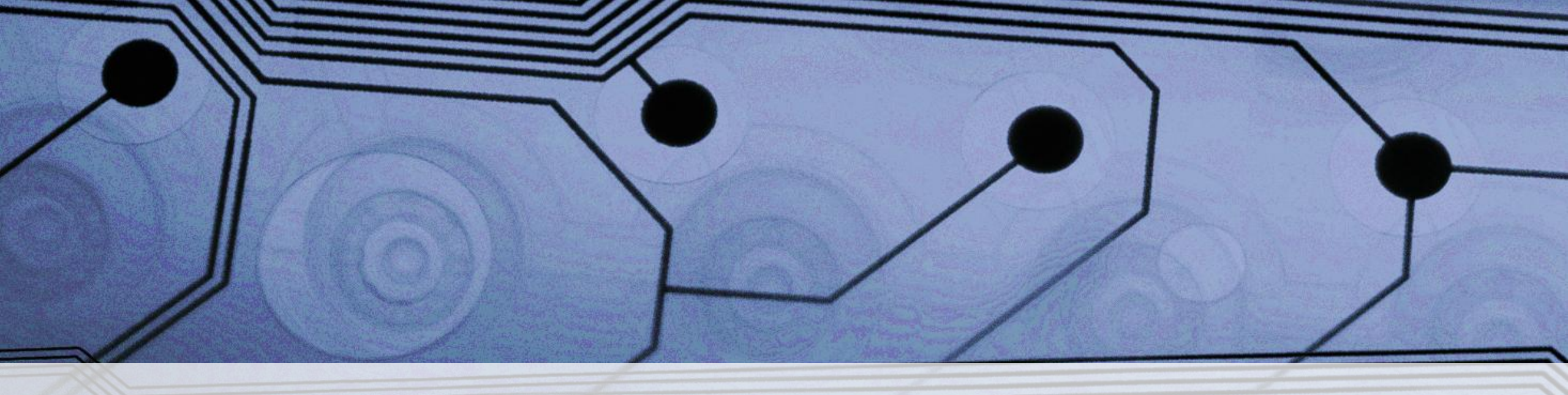
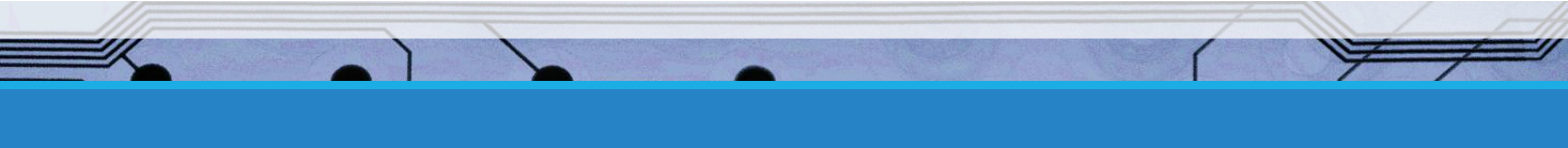




Hands-on PCB Design and Electronic Prototyping with Autodesk Fusion

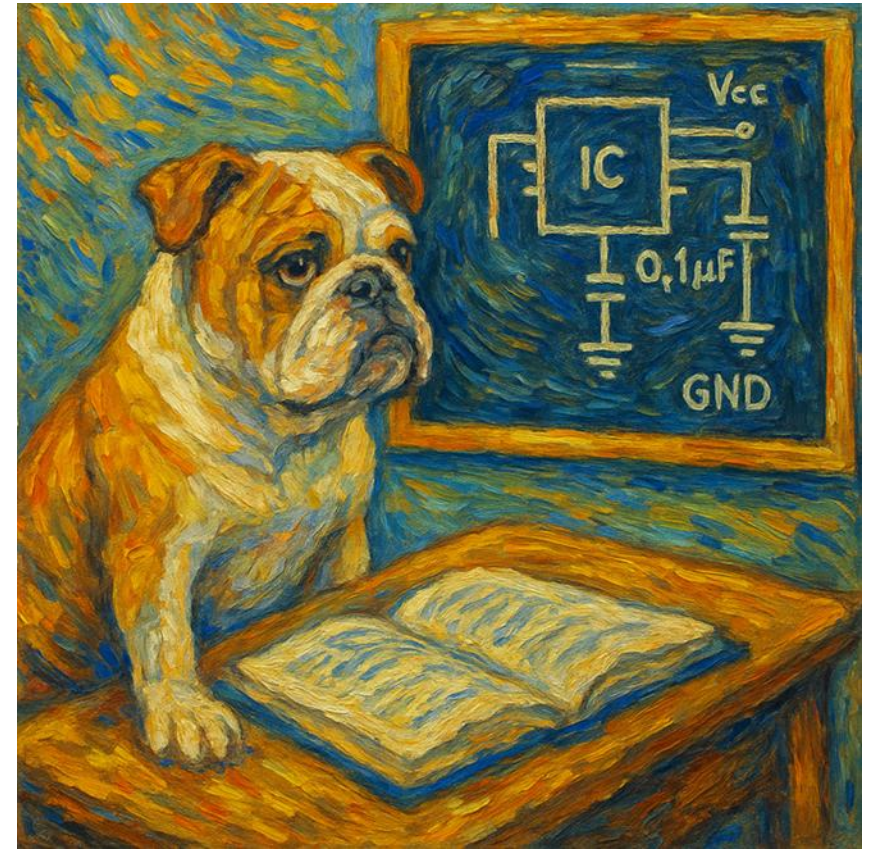
SUMMER ENRICHMENT- LYLES COLLEGE OF ENGINEERING

REVISION 1.0



Day 7: Review

- PCB Design Verification and Manufacturing Prep
 - Design Rule Checks (DRC)
 - Setting up rules and constraints
 - Identifying and correcting common errors
 - Generating Fabrication Files
 - Gerber file generation and drill files
 - Bill of Materials (BOM) and assembly documentation
 - Hands-on Activity
 - Generating manufacturing outputs for previous design
 - Reviewing Gerber and Drill files using KiCad Gerber Viewer
 - Continued work on PCB Project
 - Soldering Overview Tomorrow
 - We have some PCB's finished!!!



Day 8: Electronic Prototyping Techniques

- Rapid Prototyping Overview
 - Breadboarding best practices and limitations
 - Transitioning from breadboard to PCB prototype
- Hands-on Prototyping Skills
 - Soldering refresher: through-hole and basic SMT soldering techniques
 - Component selection and substitutions
- Hands-on Activity
 - Building and verifying circuit prototypes on breadboards and protoboards
 - Continue work on project



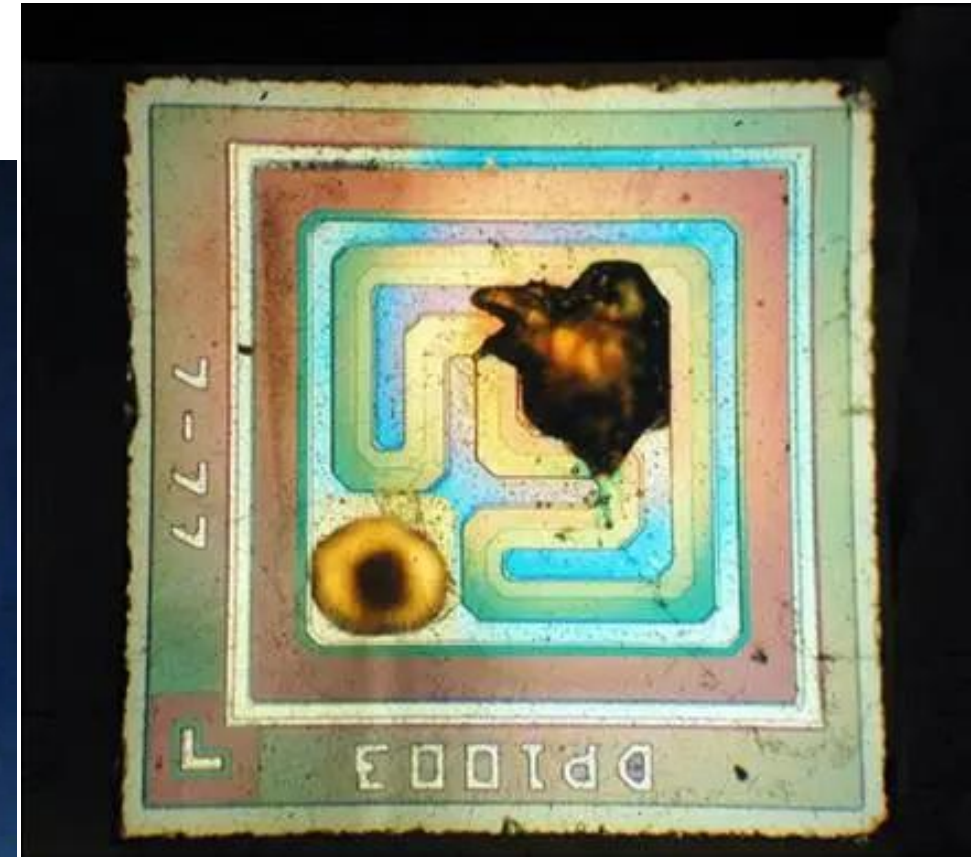
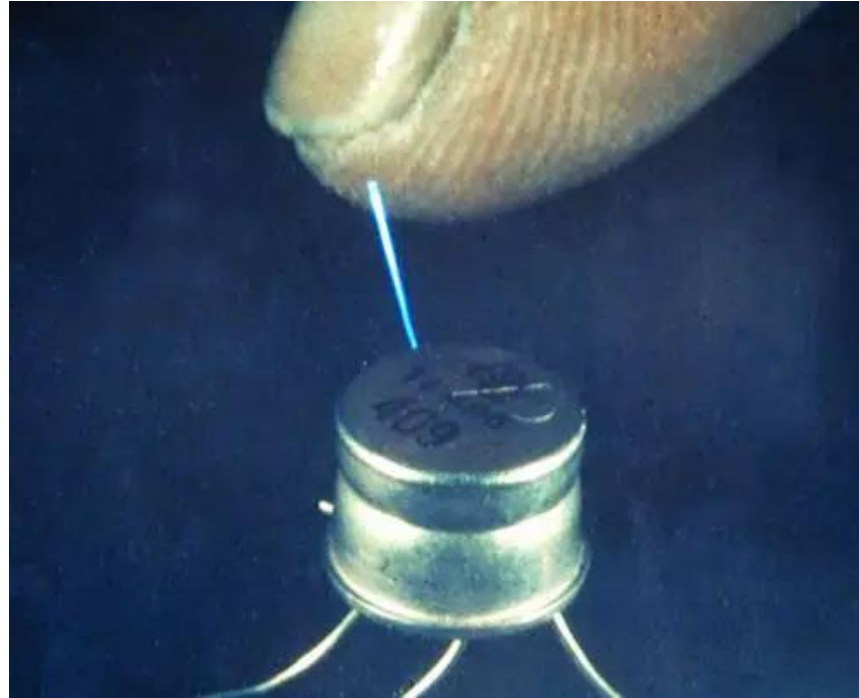
AI Minute



- Please type the following prompts into the AI Tool of your choice, Chat GPT, Gemini, Grok, Copilot, etc...
- Can you provide some solderless breadboarding best practices and solderless breadboard limitations?
-

ESD Effects on Integrated Circuits

- Photos
 - ESD arc from finger
 - Damage to silicon die

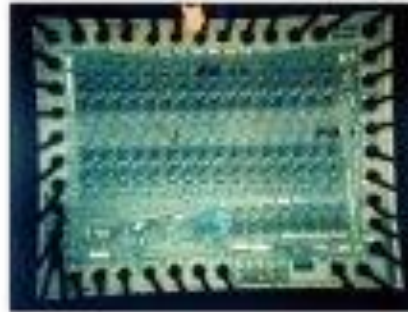


ESD Effects on Integrated Circuits

- IC Damage



ESD Damage on an Integrated Circuit (noted by arrow)



No magnification



400 x magnification



5,000 x magnification

Photos courtesy of Hi-Res Laboratories Inc., Spokane, WA

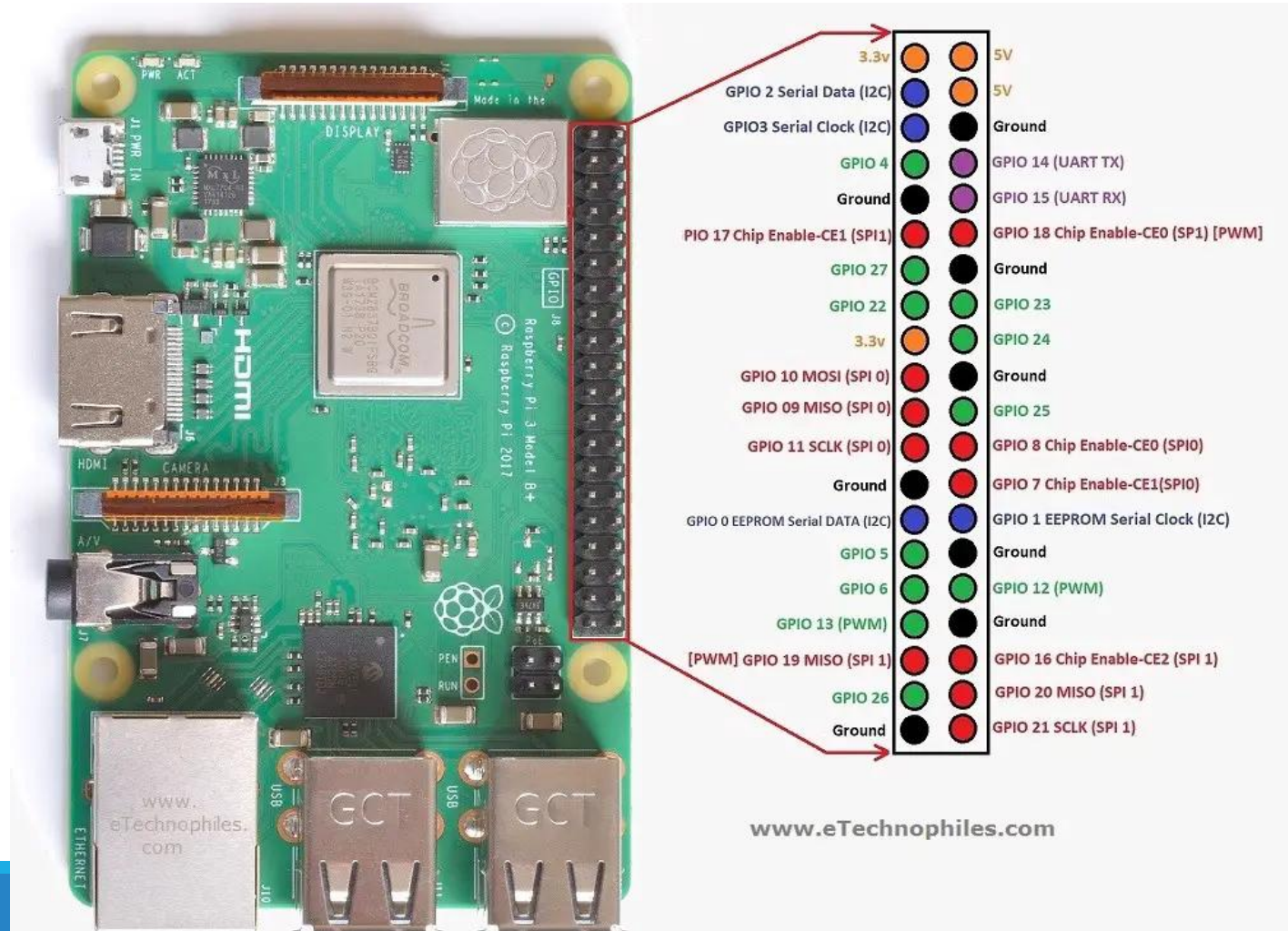
AI Minute

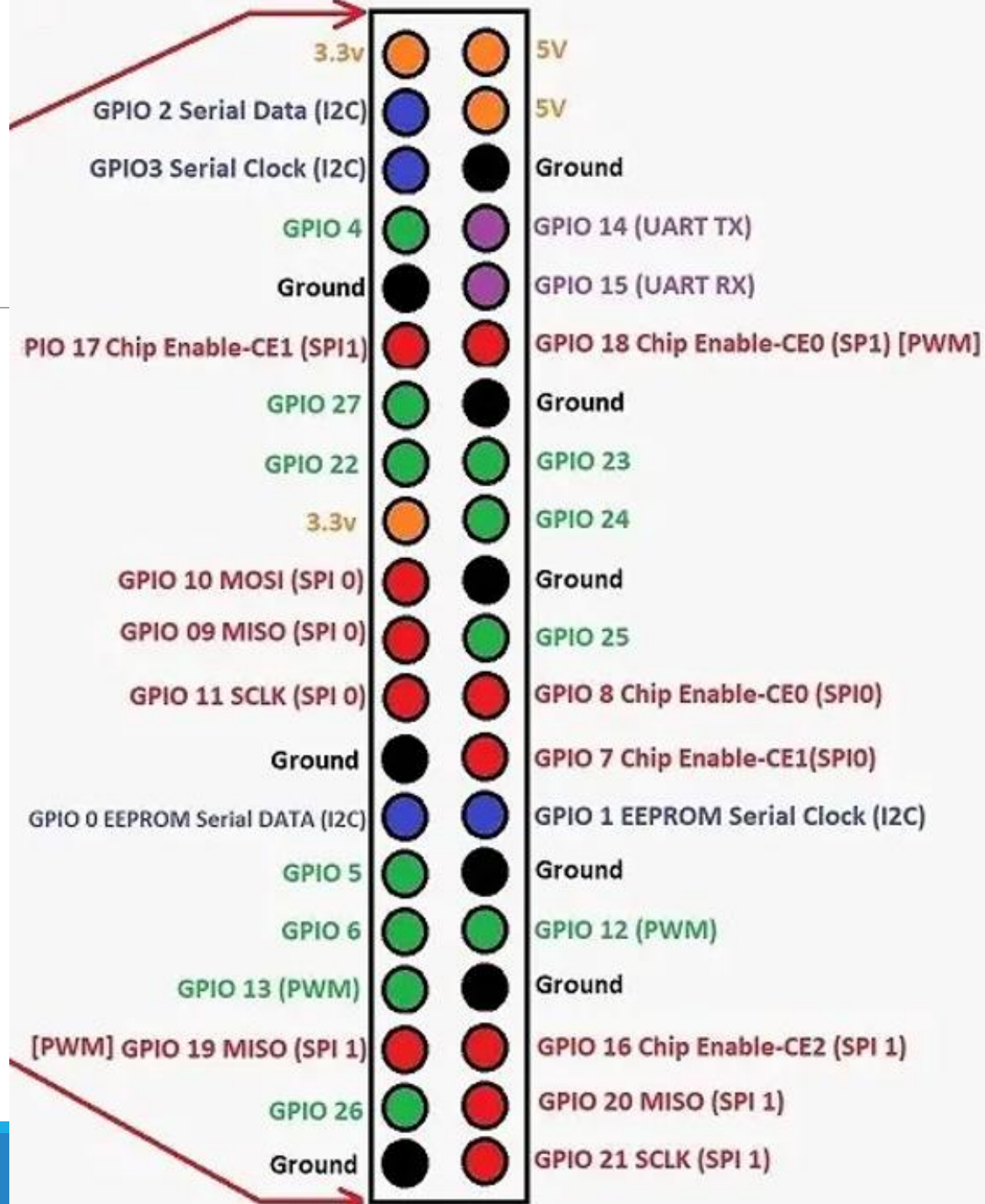


-
- Please type the following prompts into the AI Tool of your choice, Chat GPT, Gemini, Grok, Copilot, etc...
 - **How many copper atom diameters does it take to cross 1 nanometer?**

Raspberry Pi Pinout- Create Library Component- PCB Footprint

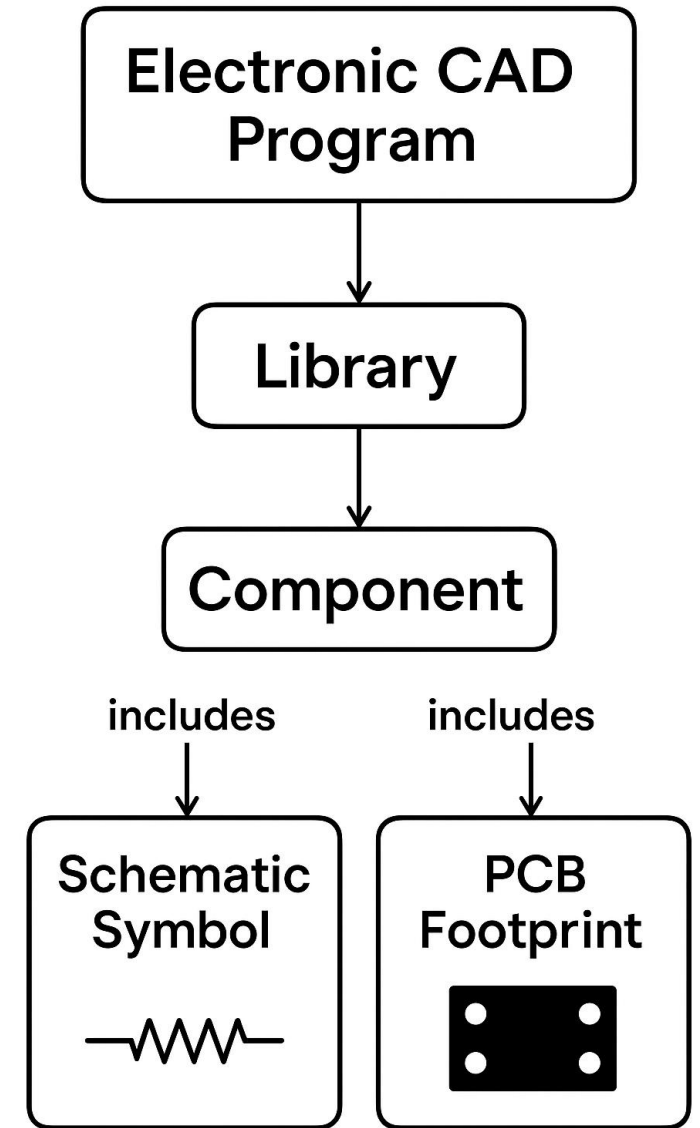
- Component Creation
 - Create Schematic Symbol
 - Create PCB Footprint
 - Combine Footprint and Symbol into a Component
- Create a new project
- Place the newly created Raspberry Pi I/O connector component on the schematic and run ERC
- Switch to PCB and place the component on the PCB
- Adjust the PCB outline to be a more appropriate size for the connector





CAD Library Elements

- Libraries
 - Libraries contain components
- Components
 - Components include:
 - A Schematic Symbol
 - A PCB Footprint
 - A 3D Model



Questions?

